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**1st Sem / ECE/ Instrumentation & Control engg. /
Automation & Robotics / Medical electronics / ECE
(For Speech and Hearing Impaired)**

Subject : Fundamental of Electrical Engg. / Med. Eltx

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Unit of voltage is

- | | |
|-----------|----------|
| a) Ampere | b) Walt |
| c) Joules | d) Volts |

Q.2 Unit of Resistance is

- | | |
|---------|----------|
| a) Ohm | b) Volt |
| c) Watt | d) Farad |

Q.3 According to Kirchoff's current law the Algebraic sum of currents meeting at a point is equal to

- | | |
|-------|-------|
| a) 1 | b) 0 |
| c) -1 | d) -2 |

a) 0 b) 2
c) 1 d) 3

a) Infinite b) 1
c) 100 d) 0

a) Ohm b) Weber
c) Farad d) Coulomb

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.8 As the temperature rises the resistance of conductor_____.

Q.10 Phase difference between voltage and current in pure resistor is equal to _____

Q.11 If Resistors of 2Ω and 3Ω are connected in series their total resistance is equal to _____.

Q.12 Reciprocal of Impedance is_____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain effect of Temperature on the Resistance of a conductor.

Q.14 Define & Explain Ohm's Law.

Q.15 Define and explain Maximum power Transfer theorem.

Q.16 Explain difference between A.C and D.C.

Q.17 Define and explain Faraday's laws of electromagnetic inductor

Q.18 Define and explain superposition theorem.

Q.19 Explain difference between primary and secondary cell.

Q.20 Write short note on solar cells and solar panel.

Q.21 Define R.M.S value of A.C.

Q.22 Define and explain Kirchhoff's Voltage law.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain construction, working principle and applications of lead Acid Batteries.

Q.24 Compare Electric circuit and Magnetic circuit.

Q.25 Explain series and parallel Resonance and its condition.